Project Overview: Cyclistic Bike-Share Case Study

Business Objective:

- *Fact:* Cyclistic's finance analysts have concluded that annual members are much more profitable than casual riders.
- *Fact:* Casual riders are already aware of the Cyclistic program and have chosen Cyclistic for their mobility needs.
- *Ultimate Goal:* Maximizing annual memberships as it's more profitable than casual rides. Converting existing casual riders into annual memberships will be cheaper, easier, and more effective than spending more to acquire completely new customers.

Stakeholders:

- Lily Moreno, Director of Marketing: She's my manager and she needs insights for her next marketing campaign (email, social media, etc). She needs actionable, realistic and marketing focused recommendations.
- Cyclistic Executive Team: They are the ultimate decision makers who are, as per the briefing document, "Notoriously Detail Oriented".

Questions to be answered:

1. *How do annual members and casual riders use Cyclistic bikes differently?*
2. *Why would casual riders buy Cyclistic annual memberships?*
3. *How can Cyclistic use digital media to influence casual riders to become members?*

Business Task:

To analyze Cyclistic's last year's bike trip data in order to identify trends, patterns and differences of how casual and membership riders use the service differently. The insights will be used by the Marketing Team for targeted strategies toward casual riders convincing them to be the highly profitable annual members.

Data Sources:

- *For this analysis, I am using Cyclistic's last 12 months' historical trip data. The data is public and made available by Motivate International Inc. under an open license.*
- *The data is stored in CSV format with a separate file for each month. The data consists variables such as ride id, bike type, start/end times, station names, and whether the rider was a casual or an annual membership rider.*
- *The data meets the ROCCC criteria as it is first party data provided directly by the company's internal systems. Due to data privacy all of the Personally Identifiable Information (PIIs) like the user's name, address, phone number or credit card details have been removed such that no individuals can be tracked. The aggregated data is sufficient enough to analyze broader patterns of different user types or classes.*

Data Processing & Cleaning:

- *I used Python (Pandas Library) to process and clean the data as the dataset contained more than 5 million rows making spreadsheet software inefficient.*
- *Data Transformation: I merged all 12 monthly datasets into a single dataframe. I converted the start and end time strings to datetime objects. I then created three new columns, ride_length (in minutes), day_of_week & month for further trend analysis.*
- *Data Cleaning: I removed duplicate records and rows with missing values(nulls) to maintain data integrity. Furthermore, I filtered out invalid entries such as rides which are of length less than a minute (likely false starts) and length more than a day (likely unreturned or stolen bikes).*

Summary of Analysis (Key Findings):

- ***Finding 1: Ride Duration & weekday trends.*** *Casual riders take significantly longer rides on average compared to the members. Members tend to only take short and purposeful rides. Members take consistent rides on weekdays (Monday to Friday) with a dip on weekends indicating they use the rides for commuting to work. On the other hand, casual members take rides usually on the weekends (Saturday and Sunday) with a dip on the weekdays clearly indicating that they use their rides for leisure and tour use.*
- ***Finding 2: Geospatial trends.*** *Usually, member stations are in financial/business districts (commutes), while casual stations are near tourist attractions, parks, or coastlines.*
- ***Finding 3: Seasonality.*** *Both user groups have significant ridership in the summers. The casual rider group is more prone to seasonal change as their ridership significantly drops in the winter months compared to the members where they show a more consistent ridership throughout the year.*

Data Visualizations:

Chart 1 - The Duration Difference

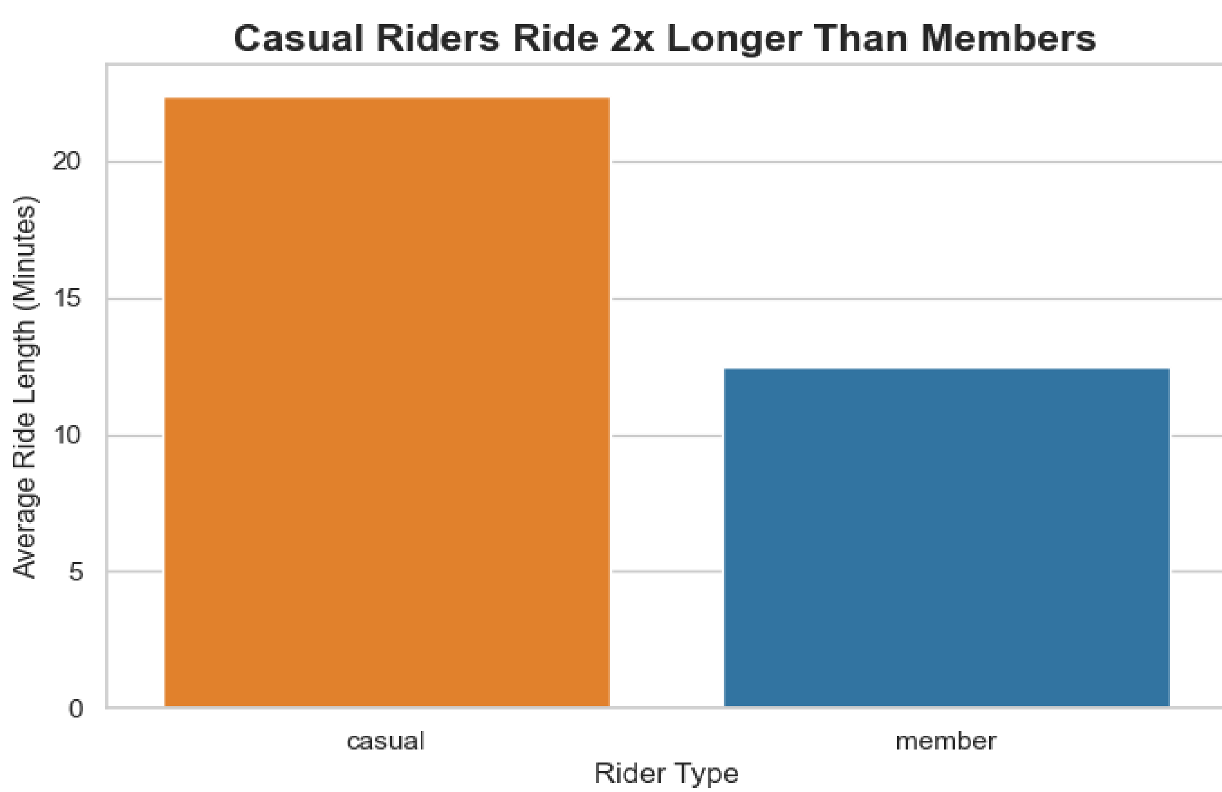
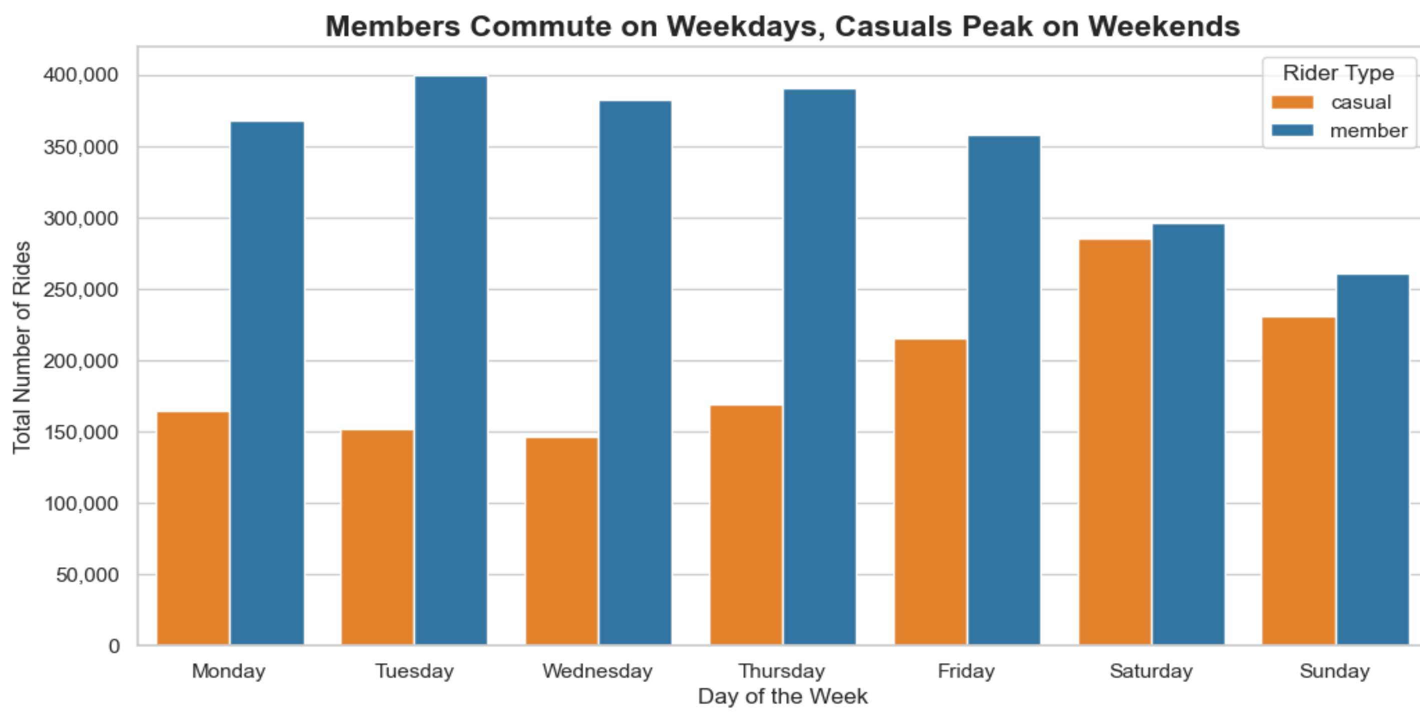
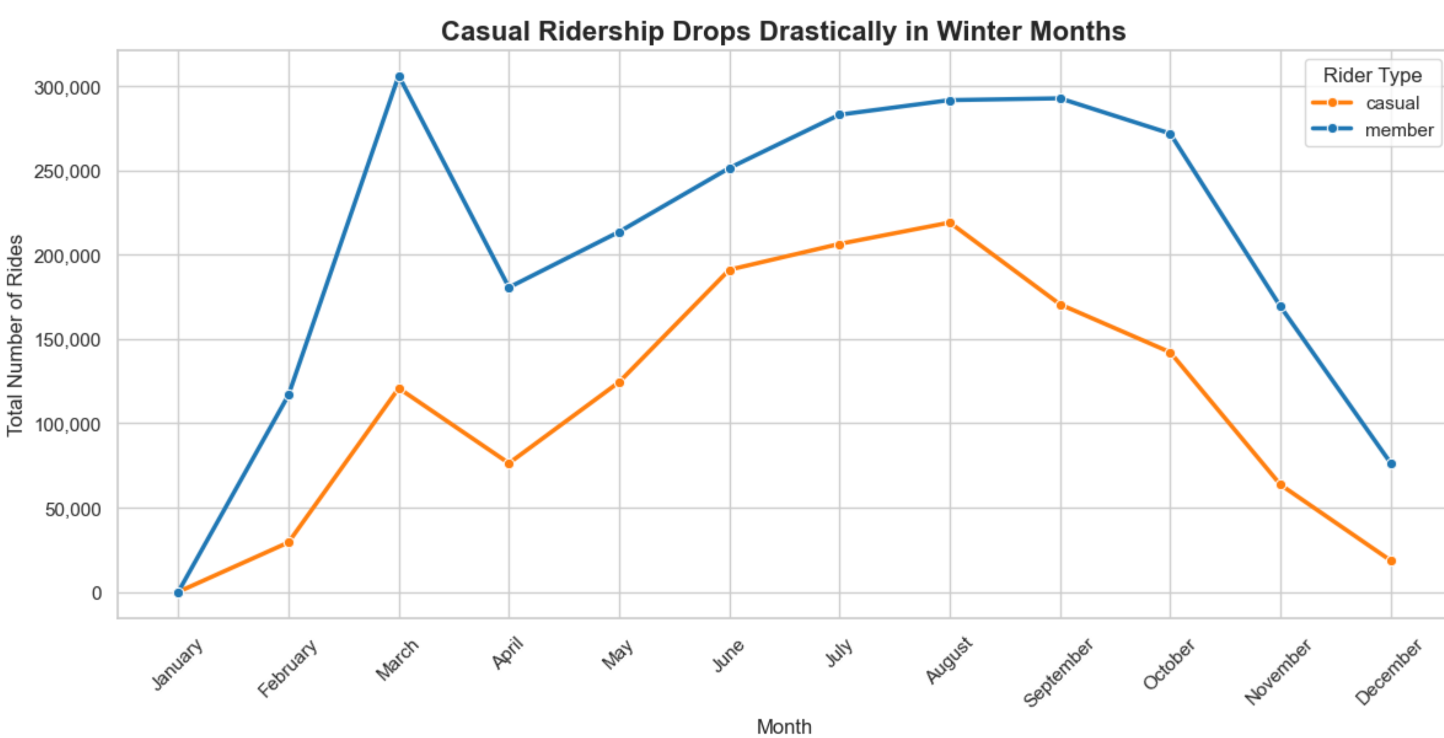


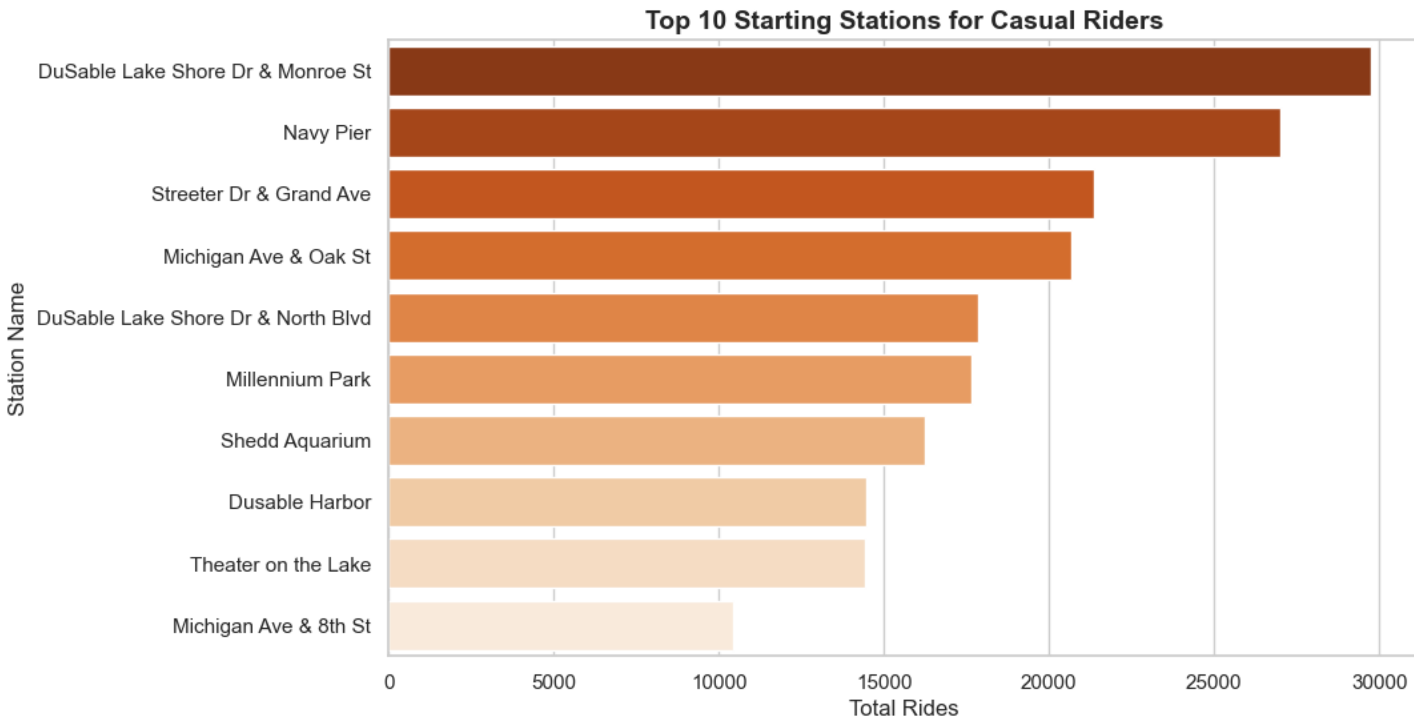
Chart 2 - The Commuter vs. Weekend Warrior



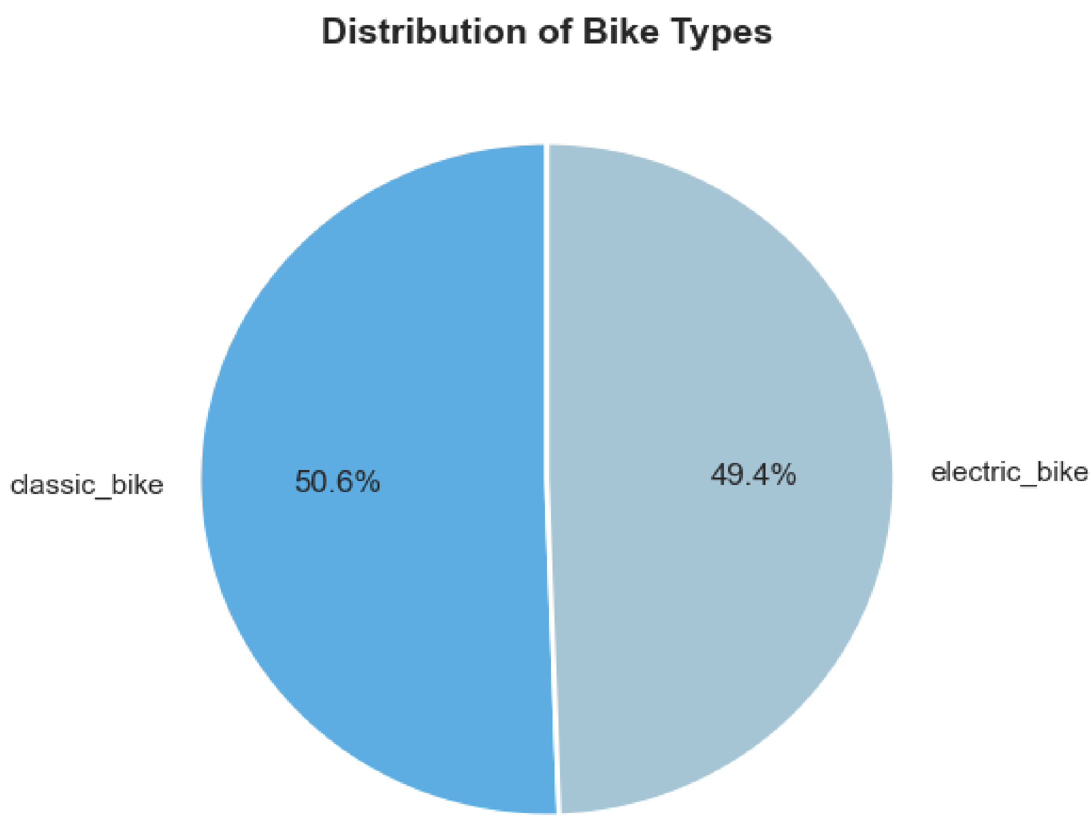
- ***Chart 3 – Seasonality***



- ***Chart 4 – Top spots for casual riders***



- ***Chart 5 – Bikes used by casuals***



Top 3 Strategic Recommendations

1. **Introduce a "Weekend-Only" Annual Membership:** Since casual riders predominantly use the service on Saturdays and Sundays, a standard 7-day annual membership may seem unnecessary to them. Offering a slightly discounted "Weekend-Only" tier provides a highly targeted stepping-stone to convert these weekend warriors into recurring members.

2. **Launch a "Summer Sundays Sale" Marketing Campaign:** Casual ridership skyrockets between June and August. Lily Moreno's team should concentrate their digital marketing budget during late Spring and early Summer. Campaigns should be placed at popular leisure destinations (parks, museums, coastlines) highlighting the special sale and discounts on the annual membership during this period.

3. **Offer "Duration-Based" Membership Perks:** Our data shows casual riders take significantly longer trips than existing members. Cyclistic could incentivize conversion by offering extended ride times (e.g., up to 60 or 90 minutes before extra fees apply) as a premium perk exclusively for annual members, directly appealing to the casual rider's behavior.

N.B.: All of the strategies should be targeted based on geospatial and time analysis to get the most R.O.I. from our expenditure.

Why would casual riders buy memberships?

1. **The "Cost of Duration" Argument**
   - **The Data:** We already proved that casual riders ride for 2x longer per trip than members.
   - **The "Why":** Most bike-shares charge casuals a base unlock fee plus a per-minute rate. If casuals are riding for 45+ minutes, they are likely paying a small fortune per ride!
   - **The Answer:** Casual riders would buy a membership if they realized that the flat annual fee is much cheaper than paying per minute for their long weekend rides.
2. **The "Location" Argument**
   - **The Data:** Usually, member stations are in financial/business districts (commutes), while casual stations are near tourist attractions, parks, or coastlines.
   - **The Answer:** Casuals would buy memberships if the membership offered exclusive parking or guaranteed bike availability at these highly congested, high-demand leisure stations.

How to use Digital Media?

1. Location-Based Digital Ads (Geofencing):

- *The Strategy:* Use digital media (Instagram/Facebook ads) to set up "geofencing" around the Top 10 casual rider stations. When a casual rider opens their phone near these locations, they see an ad for Cyclistic.
- *The Message:* "Riding near the park today? Skip the per-minute fees. Get a weekend pass!"

2. Timing-Based Email & Push Notifications:

- *The Strategy:* Since we know casual ridership spikes on Saturday and Sunday, digital marketing shouldn't be wasted on Mondays.
- *The Message:* Send targeted push notifications via the Cyclistic app on **Thursday evenings and Friday mornings**, planting the seed for their weekend plans.

3. The "Summer Campaign" Influencer Push:

- *The Strategy:* Since casual riders practically disappear in winter, digital media spend should be highly aggressive starting in May.
- *The Message:* Partner with local Chicago travel and lifestyle influencers on TikTok/Instagram to promote the health and eco-friendly benefits of a Cyclistic membership for the upcoming summer.

Future Next Steps: Additional Data to Collect

While this historical trip data provided excellent behavioral insights, to fully optimize the conversion strategy, I recommend Cyclistic collects the following additional data:

1. **Pricing & Transaction Data:** To calculate the exact dollar amount casual riders could save by switching to a membership.

2. **User Surveys:** Sending a quick email survey to casual riders asking exactly why they hesitate to buy a membership (e.g., price, lack of commitment).

3. **Weather Data:** To mathematically correlate temperature and precipitation with ridership drops, optimizing exact days to run digital ads.